

Counting matroids on ten elements

by aggregating extensions

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Abstract

We give an exact algorithm for counting matroid isomorphism classes using catalogues of smaller matroids. For each matroid and an automorphism, the algorithm counts invariant modular cuts by branching on flats below the hyperplanes and counting independent sets on the remaining hyperplanes. Weighted sums over the smaller matroids recover the Burnside terms for permutations with a fixed point; formulas expressing basis exchange supply the remaining terms. We count sparse paving matroids separately using Johnson graphs. We give pseudocode and correctness proofs for these reductions and their composition. The implementation returns 3 222 227 959 444 classes of rank five on ten elements and 3 232 000 741 644 classes across all ranks.

1 The problem and the algorithm

Let $u_{n,r}$ be the number of matroids of rank r on n elements up to isomorphism, and let $L_{n,r}$ be the number on the labeled set $[n] = \{1, \dots, n\}$. All matroids are included: loops, parallel elements, coloops, and disconnected matroids are allowed. Isomorphism means a permutation of the ground set. Duality gives $u_{n,r} = u_{n,n-r}$, but distinct duals remain distinct classes.

An extension M by one element e has a *parent* $N = M \setminus e$. The algorithm generates these smaller matroids and counts their extensions in aggregate. Its two main reductions are

one parent and one automorphism $\longrightarrow$ branching on flats $\longrightarrow$ independent sets,
counts of extensions $\longrightarrow$ sums over parents $\longrightarrow$ isomorphism classes.

The first reduction avoids constructing the final matroids. The second avoids a separate isomorphism test for each possible extension.

The main computation is rank five on ten elements. Sparse paving matroids make up most of the classes in the resulting count, and their description by independent sets of a Johnson graph gives a more efficient procedure than the general extension constraints. We therefore count them separately and restrict the general procedure to the remaining matroids:

$$u_{10,5} = u_{10,5}^{\text{sp}} + u_{10,5}^{\text{nsp}} = 2\,630\,337\,889\,305 + 591\,890\,070\,139. \quad (1)$$

The full algorithm also computes ranks three and four, uses elementary formulas for ranks zero through two, and obtains the remaining ranks by duality. Published counts are used for comparison, not as inputs.

The mathematical inputs are sets $\mathcal{P}_{q,r}$ containing one representative of each isomorphism class of matroids of rank r on q elements, with their automorphism groups. The implementation generates the catalogues from the empty matroid using the established extension generator of [4, 7]; the counting procedure computes their automorphism groups from the rank functions. The generator

produces $|\mathcal{P}_{9,5}| = 190\,214$, agreeing with [2]. The correspondence between modular cuts and extensions is due to Crapo [1]; canonical construction paths are standard [3]. The subject of this note is the aggregate counting algorithm built on these ingredients.

Notation. We use standard matroid terminology as in [5, 2]. For a matroid N , write r_N , cl_N , $\mathcal{F}(N)$, and $\text{Aut}(N)$ for its rank function, closure operator, set of flats, and automorphism group. Superscripts sp and nsp restrict a count to sparse paving matroids and matroids that are not sparse paving, respectively. The counting symbols used below are local to this note and are summarized here; each auxiliary quantity is defined in its section.

Symbol	Meaning
$u_{n,r}, L_{n,r}$	Numbers of unlabeled and labeled matroids
$F_{n,r}(\lambda)$	Number of labeled matroids fixed by a permutation of cycle type λ
$T_{n,r}(\mu)$	Weighted sum over parents for a cycle type on $n - 1$ elements
$c(N, g)$	Number of nonempty modular cuts of N invariant under g
$i(G), i_G(S)$	Numbers of independent sets of G and $G[S]$, including the empty set
$G_{N,g}$	Graph of conflicts between hyperplane orbits of a parent
Q_σ	Graph of conflicts between orbits of subsets in a Johnson graph
$E(T)$	Number of independent sets of $J(9, 4)$ containing T

2 Counting extensions of one parent

Let N be a matroid on E , of rank $r = r_N(E)$. A flat is a set equal to its closure; a hyperplane is a flat of rank $r - 1$. Two flats form a modular pair when

$$r_N(F) + r_N(H) = r_N(F \cup H) + r_N(F \cap H). \quad (2)$$

Let $e \notin E$ be the element to be added.

2.1 The Boolean formulation

A modular cut $\mathcal{C} \subseteq \mathcal{F}(N)$ is upward closed and closed under intersection of modular pairs. Crapo's correspondence [1], in the form of [5, Theorem 7.2.3], identifies it with a unique extension M satisfying $M \setminus e = N$. In this correspondence,

$$F \in \mathcal{C} \iff e \in \text{cl}_M(F). \quad (3)$$

In particular, for $X \subseteq E$,

$$r_M(X) = r_N(X), \quad r_M(X \cup \{e\}) = r_N(X) + 1 - \mathbf{1}_{\{\text{cl}_N(X) \in \mathcal{C}\}}. \quad (4)$$

The empty cut gives a coloop extension and raises the rank by one. Every nonempty cut contains E and gives a rank-preserving extension.

Introduce $x_F \in \{0, 1\}$ for every flat F . The nonempty cuts are exactly the satisfying assignments of

$$x_E = 1, \quad (5)$$

$$\neg x_F \vee x_H \quad (F \subsetneq H), \quad (6)$$

$$\neg x_F \vee \neg x_H \vee x_{F \cap H} \quad (F, H \text{ a modular pair}). \quad (7)$$

Thus counting extensions means counting these assignments, with the parent labels fixed.

For $g \in \text{Aut}(N)$, require the cut to be g -invariant by identifying flat variables in the same orbit under $\langle g \rangle$. Simplify repeated literals and delete tautological clauses. Denote the resulting formula by $\Phi_{N,g}$. Its orbit variables retain a well-defined flat rank.

Lemma 2.1. *The models of $\Phi_{N,g}$ are in bijection with rank-preserving extensions of N fixed by the permutation that acts as g on E and fixes e .*

Proof. An assignment to orbit variables is exactly a g -invariant assignment to flats. Equations (5)–(7) say precisely that its selected flats form a nonempty modular cut. The extension correspondence is equivariant: equation (4) is preserved by g exactly when the cut is preserved. Distinct cuts give distinct extensions. $\square$

2.2 Reduction to independent sets

In this section, call a flat *lower* if its rank is at most $r - 2$. Branch on the variables for orbits of lower flats and apply unit propagation after every assignment. Once these variables are assigned, only variables for hyperplane orbits can remain undecided.

For fixed (N, g) , define a simple graph $G_{N,g}$ whose vertices are the $\langle g \rangle$ -orbits of hyperplanes of N . Two distinct vertices O, O' are adjacent if and only if some $H \in O$ and $K \in O'$ satisfy $r_N(H \cap K) = r - 2$. We call this the *conflict graph*. An orbit can also contain two distinct hyperplanes with this property; the modular cut clauses handle such an internal conflict.

Lemma 2.2 (Hyperplane reduction). *Suppose all variables for lower flats are assigned and unit propagation has reached a consistent fixed point. Let S be the set of undecided hyperplane orbits. The satisfying completions of this assignment are in bijection with the independent sets of $G_{N,g}[S]$.*

Proof. Distinct hyperplanes span N , so (2) becomes $r_N(H \cap K) = r - 2$. Their intersection is a lower flat. If its variable is true, upward closure forces both hyperplanes true. Consequently, whenever both hyperplane variables remain undecided, their clause reduces to $\neg x_H \vee \neg x_K$.

Every other clause has at most one undecided variable: its other flats are lower or equal to E . Such a clause is either satisfied or has already been resolved by unit propagation. If two conflicting hyperplanes lie in the same orbit, a true intersection forces the orbit true, while a false intersection forces it false. Such an orbit cannot remain undecided. Thus the residual clauses are exactly the edge constraints of $G_{N,g}[S]$. Selecting an independent set specifies exactly one completion of the cut. $\square$

The graph $G_{N,g}$ is built once for each (N, g) ; branching changes only the set S . A parent of rank five on nine elements has at most $\binom{9}{4} = 126$ hyperplanes and at most $\sum_{j=0}^3 \binom{9}{j} = 130$ lower flats: a rank- j flat is the closure of some basis of size j . This bounds the graph size in the implementation.

2.3 Counting independent sets

For a simple graph G , let $i(G)$ count its independent sets, including the empty set, and write $i_G(S) = i(G[S])$ for $S \subseteq V(G)$. Write $N_G(v)$ for the open neighborhood of v . Algorithm 1 uses a cache keyed by the complete set S for a fixed graph. The cache may discard an entry whenever space is needed.

Algorithm 1. $\text{IS}_G(S)$

1. If $S = \emptyset$, return 1. If S is cached, return its value.
 2. Save $S_0 \leftarrow S$. Remove all isolated vertices of $G[S]$; let k be their number and S the remaining set.
 3. If $S = \emptyset$, set $a \leftarrow 1$.
 4. Otherwise, if component splitting finds a component $C \subsetneq S$, set $a \leftarrow \text{IS}_G(C) \text{IS}_G(S \setminus C)$.
 5. Otherwise, if $G[S]$ is a path on t vertices, set $a \leftarrow f_{t+2}$; if it is a cycle, set $a \leftarrow f_{t-1} + f_{t+1}$.
 6. Otherwise choose $v \in S$ of maximum degree and set $a \leftarrow \text{IS}_G(S \setminus \{v\}) + \text{IS}_G(S \setminus (\{v\} \cup N_G(v)))$.
 7. Store $2^k a$ under the key S_0 , discarding a cache entry if needed, and return $2^k a$.
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Here $f_0 = 0, f_1 = 1$ and $f_{t+2} = f_{t+1} + f_t$ define the Fibonacci numbers. The rule for deciding when to search for components is an implementation choice described in Section 7.1.

Proposition 2.3. *Algorithm 1 returns $i_G(S)$.*

Proof. An isolated vertex can be independently included or excluded, giving 2^k . Choices on distinct components are independent, so their counts multiply. Partitioning by whether v is selected gives

$$i_G(S) = i_G(S \setminus \{v\}) + i_G(S \setminus (\{v\} \cup N_G(v))).$$

For paths, the same recurrence at an endpoint gives f_{t+2} ; conditioning on one vertex of a cycle gives $f_{t+1} + f_{t-1}$. Every recursive argument is smaller, proving termination and correctness by induction on $|S|$. Cache entries store previously established values of the same function. Discarding an entry changes only the amount of repeated work. $\square$

2.4 Counting modular cuts

We allow a set $\mathcal{B}$ of *marked* orbit variables and a flag b requiring at least one of them to be true. Ordinary extension counting uses $b = 0$. Section 4 specifies the marks used to exclude sparse paving extensions. Build $\Phi_{N,g}$ and $G = G_{N,g}$ once, and start with $x_E = 1$.

Algorithm 2. $\text{Cut}(\alpha, b)$ for fixed $\Phi_{N,g}$, $G = G_{N,g}$, and $\mathcal{B}$

1. Propagate unit clauses from the partial assignment α . If a contradiction occurs, return 0.
 2. If a marked variable is true, set $b \leftarrow 0$.
 3. If an undecided lower variable v exists, return $\text{Cut}(\alpha \cup \{v = 1\}, b) + \text{Cut}(\alpha \cup \{v = 0\}, b)$, using a separate state for each branch.
 4. Let S be the set of undecided hyperplane variables.
 5. If $b = 0$, return $\text{IS}_G(S)$.
 6. Set $a \leftarrow 0$. List the marked vertices in S as $v_1, \dots, v_t$.
 7. For $j = 1, \dots, t$: set $S \leftarrow S \setminus \{v_j\}$ and $a \leftarrow a + \text{IS}_G(S \setminus N_G(v_j))$.
 8. Return a .
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The cache for IS_G is fresh for each (N, g) , since a key consisting only of S has meaning only after the graph G is fixed.

Proposition 2.4 (Correctness of the extension count). *Algorithm 2 counts exactly the models extending α , subject to the requirement that some marked variable is true when $b = 1$.*

Proof. Unit propagation forces values shared by every satisfying completion. A contradiction gives no completions. Branching on v partitions all completions into two disjoint sets and decreases the number of undecided lower variables. At a leaf, Lemma 2.2 and Proposition 2.3 give the unrestricted count.

If a marked variable must still be selected, partition the independent sets by their first selected vertex in $v_1, \dots, v_t$. For the j th part, $v_1, \dots, v_{j-1}$ are excluded and v_j is included. The remaining choices are precisely the independent sets of $G[S_0 \setminus (\{v_1, \dots, v_j\} \cup N_G(v_j))]$, where S_0 was the initial leaf set. These are the terms in lines 6–7. They are disjoint and exhaust the required models; when $t = 0$ their sum is zero. $\square$

Example 2.5 (Five extensions of a three-point line). Take $N = U_{2,3}$. Its flats are $\emptyset$, the three singletons, and E . If $x_\emptyset = 1$, every flat is selected: the new element is a loop. If $x_\emptyset = 0$ and g is the identity, the conflict graph is K_3 , with four independent sets. The empty set gives $U_{2,4}$, and each singleton makes e parallel to one existing point. Thus there are $1 + 4 = 5$ rank-preserving labeled extensions. For the automorphism (12), the orbit of the first two hyperplanes has an internal conflict and is excluded when $x_\emptyset = 0$. Only the empty choice and the third hyperplane remain, giving $1 + 2 = 3$ invariant extensions. The coloop extension is separate and has rank three.

3 From parent counts to isomorphism classes

Let $F_{n,r}(\lambda)$ be the number of labeled matroids of rank r fixed by one permutation of cycle type $\lambda \vdash n$. If $a_j(\lambda)$ is the number of parts equal to j , set

$$z_\lambda = \prod_{j \geq 1} j^{a_j(\lambda)} a_j(\lambda)!$$

There are $n!/z_\lambda$ permutations of this type. Burnside's lemma gives

$$u_{n,r} = \sum_{\lambda \vdash n} \frac{F_{n,r}(\lambda)}{z_\lambda}. \quad (8)$$

The identity term $F_{n,r}(1^n)$ is $L_{n,r}$.

3.1 Permutations with a fixed element

Put $q = n - 1$. For $N \in \mathcal{P}_{q,r}$ and $g \in \text{Aut}(N)$, let $c(N, g)$ be the number of nonempty modular cuts invariant under g , computed by Algorithm 2 with $b = 0$. For every $\mu \vdash q$, define the following weighted sum over parents:

$$\begin{aligned} T_{n,r}(\mu) = & \sum_{N \in \mathcal{P}_{q,r}} \frac{q!}{|\text{Aut}(N)|} \sum_{\substack{g \in \text{Aut}(N) \\ \text{type}(g) = \mu}} c(N, g) \\ & + \sum_{N \in \mathcal{P}_{q,r-1}} \frac{q!}{|\text{Aut}(N)|} \#\{g \in \text{Aut}(N) : \text{type}(g) = \mu\}. \end{aligned} \quad (9)$$

The second line counts coloop extensions. Write $\mu \cup (1)$ for adjoining one part of size one.

Theorem 3.1 (Counting by deletion). *For every $\mu \vdash n - 1$,*

$$F_{n,r}(\mu \cup (1)) = \frac{z_\mu T_{n,r}(\mu)}{(n-1)!}. \quad (10)$$

Proof. Fix the added label $e = n$. Count pairs (M, g) , where M is a labeled matroid of rank r on $[n]$, g is a permutation of $[q]$ of type μ , and g extended by fixing e preserves M .

Delete e . If e is not a coloop, the parent has rank r , is preserved by g , and Lemma 2.1 supplies $c(N, g)$ choices. If e is a coloop, the parent has rank $r - 1$ and there is exactly one extension. Each parent type has $q! / |\text{Aut}(N)|$ labelings. Summing by parent therefore gives precisely (9).

Alternatively, there are $q! / z_\mu$ choices of g . They are conjugate in S_q , so each fixes $F_{n,r}(\mu \cup (1))$ matroids. Equating the two counts proves (10). $\square$

The contribution to (8) from permutations with a fixed point is

$$\sum_{\mu \vdash q} \frac{T_{n,r}(\mu)}{q! (a_1(\mu) + 1)}, \quad (11)$$

because $z_{\mu \cup (1)} = (a_1(\mu) + 1) z_\mu$. This factor is the precise correction for adding a distinguished fixed element. Summing extension orbits directly would instead count pointed matroids (M, e) , with one occurrence per element orbit of M .

Conjugacy classes. If $g' = hgh^{-1}$ for $h \in \text{Aut}(N)$, the map $\mathcal{C} \mapsto h(\mathcal{C})$ bijects the invariant cuts for g and g' . Thus each inner sum in (9) may be evaluated using one representative per conjugacy class of $\text{Aut}(N)$, multiplied by its class size. Equal cycle types in S_q alone do not establish this bijection for a fixed parent: the conjugating permutation must preserve N .

3.2 Permutations without a fixed element

For a representative σ of type $\lambda \vdash n$, introduce one Boolean variable y_O for every orbit O of r -subsets under $\langle \sigma \rangle$. Let $[B]$ denote the orbit of B . Use the formula

$$\Psi_{n,r,\sigma} = \left(\bigvee_O y_O \right) \wedge \bigwedge_{\substack{B, C \in \binom{[n]}{r} \\ a \in B \setminus C}} \left(\neg y_{[B]} \vee \neg y_{[C]} \vee \bigvee_{b \in C \setminus B} y_{[(B \setminus \{a\}) \cup \{b\}]} \right). \quad (12)$$

Write $\#\text{MC}(\Psi)$ for its exact number of satisfying assignments.

Proposition 3.2. $\#\text{MC}(\Psi_{n,r,\sigma}) = F_{n,r}(\lambda)$.

Proof. An assignment selects a unique σ -invariant family of r -subsets. The first clause makes it nonempty. The remaining clauses are exactly the basis exchange axiom, which characterizes the bases of matroids of rank r [5, Section 1.2]. Conversely, the bases of every fixed matroid give one satisfying assignment. No auxiliary variables occur, so the correspondence preserves the number of models without any projection factor. $\square$

For a hard formula we choose k variables and count all 2^k restrictions. Every model has exactly one assignment to these variables, hence

$$\#MC(\Psi) = \sum_{\eta \in \{0,1\}^k} \#MC(\Psi \wedge \eta). \quad (13)$$

These jobs can be scheduled independently. At $n = 10$ there are 42 cycle types: 30 with a fixed point and 12 without one. The latter are the only types for which we count models of $\Psi_{n,r,\sigma}$.

Algorithm 3. Full rank count $\text{RankCount}(n, r)$, $1 \leq r < n$

1. Set $q \leftarrow n - 1$ and $T_{n,r}(\mu) \leftarrow 0$ for every $\mu \vdash q$.
 2. For each $N \in \mathcal{P}_{q,r}$, put $w \leftarrow q! / |\text{Aut}(N)|$. For every representative g of a conjugacy class in $\text{Aut}(N)$ with class size s , compute $c \leftarrow \text{Cut}(x_E = 1, 0)$ and add wsc to $T_{n,r}(\text{type}(g))$.
 3. For each $N \in \mathcal{P}_{q,r-1}$ and each such class (s, g) , add $q!s / |\text{Aut}(N)|$ to $T_{n,r}(\text{type}(g))$.
 4. For each $\mu \vdash q$, set $F_{n,r}(\mu \cup (1)) \leftarrow z_\mu T_{n,r}(\mu) / q!$, checking exact division.
 5. For each $\lambda \vdash n$ with no part 1, choose σ of that type and set $F_{n,r}(\lambda) \leftarrow \#MC(\Psi_{n,r,\sigma})$.
 6. Set $A \leftarrow \sum_{\lambda \vdash n} (n! / z_\lambda) F_{n,r}(\lambda)$. Return $(A/n!, F_{n,r}(1^n))$, checking exact division.
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Theorem 3.3. *Given complete parent catalogues and their automorphism groups, Algorithm 3, with exact formula counts, returns $(u_{n,r}, L_{n,r})$.*

Proof. Proposition 2.4 supplies the correct invariant cut counts. Summing over conjugacy classes gives the inner sums of (9). Theorem 3.1 supplies every term for a permutation with a fixed point, and Proposition 3.2 supplies every remaining term. These two sets of cycle types partition all of S_n . Equation (8) proves the unlabeled count; the identity fixes every labeled matroid and gives the second output. $\square$

4 Excluding sparse paving extensions

A matroid of rank r is paving if every circuit has size at least r , and sparse paving if both it and its dual are paving. For rank five on ten elements this means that every circuit and every cocircuit has size at least five. Equivalently, it is paving and every hyperplane has at most five elements, since the complement of a hyperplane is a cocircuit.

In this section $N = M \setminus e$ has rank five on nine elements, and M also has rank five. If N is not sparse paving, neither is M : a circuit of N of size at most four is also a circuit of M , and a hyperplane of N of size at least six lies in a hyperplane of M of size at least six.

Lemma 4.1 (The marked flats). *Suppose N is sparse paving and $\mathcal{C}$ is the nonempty modular cut determining M . The extension M is not sparse paving if and only if $\mathcal{C}$ contains either a flat of rank at most three or a hyperplane of size five.*

Proof. If $F \in \mathcal{C}$ has rank at most three, choose a basis I of F . Then $e \in \text{cl}_M(I)$, so $I \cup \{e\}$ contains a circuit through e of size at most four. If a hyperplane H of size five belongs to $\mathcal{C}$, its closure in M is $H \cup \{e\}$, a hyperplane of size six. Its complement is a cocircuit of size four. Either case violates sparse paving.

Conversely, if M has a circuit D of size at most four, then $e \in D$ because N is paving. The flat $\text{cl}_N(D \setminus \{e\})$ has rank at most three and belongs to $\mathcal{C}$. Otherwise M is paving, so failure of sparse paving gives a hyperplane K with $|K| \geq 6$. It must contain e : a hyperplane of M avoiding e is a hyperplane of N , whose size is at most five. The flat $H = K \setminus \{e\}$ of N has at least five elements. Since every subset of N of size four is independent, $r_N(H) = 4$. Thus H is a hyperplane, $|H| = 5$, and $r_M(H \cup \{e\}) = r_N(H)$, proving $H \in \mathcal{C}$. $\square$

Let $c^{\text{nsp}}(N, g)$ be the number of nonempty modular cuts invariant under g whose extensions are not sparse paving. If N is not sparse paving, use Algorithm 2 with $b = 0$. Otherwise mark the orbit variables containing the flats in Lemma 4.1, and use $b = 1$. These marks are invariant under $\text{Aut}(N)$, so the reduction to conjugacy classes still applies. Proposition 2.4 proves that the omitted branch is exactly the sparse paving family.

4.1 The coloop contribution at the middle rank

Every coloop extension of rank five on ten elements fails to be sparse paving, because a coloop is a cocircuit of size one. Duality bijects $\mathcal{P}_{9,5}$ and $\mathcal{P}_{9,4}$ and preserves their automorphism groups as permutation groups. Consequently the two parent sums can be paired:

$$T_{10,5}^{\text{nsp}}(\mu) = \sum_{N \in \mathcal{P}_{9,5}} \frac{9!}{|\text{Aut}(N)|} \sum_{\substack{g \in \text{Aut}(N) \\ \text{type}(g) = \mu}} (c^{\text{nsp}}(N, g) + 1), \quad \mu \vdash 9. \quad (14)$$

The $+1$ is the coloop extension of the dual parent. This pairing is used only at $(n, r) = (10, 5)$; the general rank calculation keeps the two parent sets in (9) separate.

Applying Theorem 3.1 to the family of matroids that are not sparse paving gives

$$F_{10,5}^{\text{nsp}}(\lambda) = \begin{cases} z_\mu T_{10,5}^{\text{nsp}}(\mu)/9!, & \lambda = \mu \cup (1), \\ \#\text{MC}(\Psi_{10,5,\sigma}) - F_{10,5}^{\text{sp}}(\lambda), & 1 \notin \lambda. \end{cases} \quad (15)$$

Thus

$$u_{10,5}^{\text{nsp}} = \sum_{\lambda \vdash 10} \frac{F_{10,5}^{\text{nsp}}(\lambda)}{z_\lambda}. \quad (16)$$

Here σ has cycle type λ . For permutations with a fixed point, the procedure omits all extensions in which every marked variable is false. For the twelve derangement types, it counts all matroids using basis exchange and subtracts the number that are sparse paving.

5 The sparse paving calculation

For $1 \leq r < n$, let $J(n, r)$ be the Johnson graph: its vertices are the r -subsets of $[n]$, with two vertices adjacent if and only if their intersection has size $r - 1$. The circuit-hyperplanes of a sparse paving matroid of rank r form an independent set of this graph, and every independent set specifies such a matroid [2, 6]. These sets are also called *stable sets* in the literature. Thus $L_{n,r}^{\text{sp}} = i(J(n, r))$. This description removes the need to branch on lower flats and is the reason for counting this family separately. All orbits below are under permutations of the ground set. When $n = 2r$, complementation also preserves $J(n, r)$; we do not quotient by it, because it corresponds to matroid duality.

5.1 The identity permutation

Distinguish $e = n$ and split the family of circuit-hyperplanes into A , the blocks containing e with e removed, and B , the blocks avoiding e . Define

$$\Gamma^+(A) = \left\{ b \in \binom{[n-1]}{r} : \exists a \in A, a \subset b \right\}.$$

Then A is independent in $J(n-1, r-1)$, B is independent in $J(n-1, r)$, and no block in one part is adjacent to a block in the other exactly when $B \cap \Gamma^+(A) = \emptyset$.

Proposition 5.1 (Splitting at one element). *If $\mathcal{A}$ contains one representative from every S_{n-1} -orbit of independent sets of $J(n-1, r-1)$, then*

$$L_{n,r}^{\text{sp}} = \sum_{A \in \mathcal{A}} \frac{(n-1)!}{|\text{Aut}(A)|} i(J(n-1, r) - \Gamma^+(A)). \quad (17)$$

Proof. Every family of circuit-hyperplanes has a unique pair (A, B) . For a fixed labeled A , its possible B are the independent sets of the displayed residual graph. Relabeling A gives an isomorphic residual graph, so all $(n-1)!/|\text{Aut}(A)|$ members of its orbit have the same number of extensions. Summing over the parent orbits counts every labeled family once. $\square$

5.2 Completion counts for small parents

A parent with few blocks forbids few vertices, leaving a large residual graph. These cases are expensive for direct counting. For $(n, r) = (10, 5)$, we instead use inclusion–exclusion and a table of completion counts when the parent has at most five blocks.

Complementation on $[9]$ identifies $J(9, 5)$ with $G = J(9, 4)$. A parent block $a \in \binom{[9]}{4}$ forbids its five supersets of size five. After complementation these become precisely

$$D_a = \binom{[9] \setminus a}{4},$$

the five subsets of size four of $[9] \setminus a$. They form a clique in G . Two distinct blocks of A cannot share a superset of size five, because A is independent in $J(9, 4)$. Thus the forbidden set $D = \bigcup_{a \in A} D_a$ is a union of $|A|$ disjoint cliques.

For an independent set T of G , define $E(T) = \#\{H \subseteq V(G) : H \text{ independent, } T \subseteq H\}$. Let $o(H) = 9!/|\text{Aut}(H)|$ and $m_T(H) = \#\{T' \subseteq H : T' \cong T\}$, with isomorphism under S_9 .

Proposition 5.2 (Counting completions and applying inclusion–exclusion). *Given a catalogue $\mathcal{H}$ containing one representative of every S_9 -orbit of independent sets of G , we have*

$$E(T) = \frac{\sum_{H \in \mathcal{H}} o(H) m_T(H)}{o(T)}, \quad (18)$$

$$i(G - D) = \sum_{\substack{T \subseteq D \\ T \text{ independent}}} (-1)^{|T|} E(T). \quad (19)$$

If D is covered by k cliques, all terms with $|T| > k$ vanish.

Proof. For (18), count pairs (H, T') over all labeled independent sets H , with $T' \subseteq H$ and $T' \cong T$. Summing first over H gives the numerator. Each of the $o(T)$ copies of T has the same number $E(T)$ of completions, giving the denominator.

For (19), a fixed independent set H contributes $\sum_{T \subseteq H \cap D} (-1)^{|T|}$, which equals one when $H \cap D = \emptyset$ and zero otherwise. Every such T is independent. An independent set can meet each covering clique in at most one vertex, so its size is at most k . $\square$

The implementation tabulates $E(T)$ for $|T| \leq 5$ from the freshly generated catalogue of sparse paving matroids of rank four on nine elements. This handles all 251 parent types with $|A| \leq 5$. Larger parents use Algorithm 1 on the residual graph. This is a choice between two exact formulas for the same count.

5.3 The other permutations

For $\sigma \in S_n$, an invariant family of circuit-hyperplanes is a union of $\langle \sigma \rangle$ -orbits of r -subsets. Delete any orbit containing two members that are adjacent in $J(n, r)$. Form a graph Q_σ on the remaining orbits, with two vertices adjacent if and only if some member of one orbit is adjacent in $J(n, r)$ to a member of the other. Selecting an independent set of Q_σ is equivalent to selecting a σ -invariant independent set of $J(n, r)$; hence

$$F_{n,r}^{\text{sp}}(\text{type}(\sigma)) = i(Q_\sigma). \quad (20)$$

Algorithm 4. Counting invariant sparse paving matroids at $(n, r) = (10, 5)$

1. Generate the complete catalogue $\mathcal{H}$ of independent sets of $J(9, 4)$ modulo S_9 , including their automorphism orders.
 2. Compute $E(T)$ for the types with $|T| \leq 5$ using (18).
 3. For each $A \in \mathcal{H}$, compute its residual count by (19) if $|A| \leq 5$, and by IS otherwise.
 4. Set $F_{10,5}^{\text{sp}}(1^{10})$ to the sum (17).
 5. For every other $\lambda \vdash 10$, choose a permutation σ of type λ , construct Q_σ , and set $F_{10,5}^{\text{sp}}(\lambda) \leftarrow \text{IS}_{Q_\sigma}(V(Q_\sigma))$.
 6. Return the table $F_{10,5}^{\text{sp}}$ and $u_{10,5}^{\text{sp}} = \sum_{\lambda \vdash 10} F_{10,5}^{\text{sp}}(\lambda)/z_\lambda$.
-

Corollary 5.3. *Algorithm 4 returns the numbers of fixed sparse paving matroids and their unlabeled total.*

Proof. Propositions 5.1 and 5.2 give the identity term. The bijection given by selecting orbits proves (20) for every other term. Proposition 2.3 evaluates these graph counts, and Burnside's lemma gives the final sum. $\square$

6 Counting all matroids on ten elements

For the three smallest ranks, direct formulas are sufficient. For $n \geq 2$,

$$\begin{aligned} (u_{n,0}, L_{n,0}) &= (1, 1), \\ (u_{n,1}, L_{n,1}) &= (n, 2^n - 1), \\ u_{n,2} &= \sum_{s=2}^n (p(s) - 1), \quad L_{n,2} = B_{n+1} - 2^n. \end{aligned} \tag{21}$$

Here $p(s)$ is the integer partition number and B_s is the Bell number. Rank one is determined by its nonempty set of nonloops. A matroid of rank two is determined by its loops and a partition of the nonloops into at least two parallel classes. Up to isomorphism, the class sizes are an integer partition of some $s \leq n$ with at least two parts. For labeled counting, adjoin a marker to the loop block and partition the resulting $n + 1$ elements. The forbidden cases with zero or one nonloop block number $1 + (2^n - 1) = 2^n$. This proves (21).

Algorithm 5. All matroids on ten elements

1. Generate the parent catalogues $\mathcal{P}_{9,r}$ needed below, with their automorphism groups, and the sparse paving catalogue for Algorithm 4.
 2. Run Algorithm 4 to obtain $F_{10,5}^{\text{sp}}(\lambda)$ for all $\lambda \vdash 10$.
 3. Set $T_{10,5}^{\text{nsp}}(\mu) \leftarrow 0$ for all $\mu \vdash 9$. For each $N \in \mathcal{P}_{9,5}$ and each automorphism conjugacy class (s, g) , compute $c^{\text{nsp}}(N, g)$ as in Section 4 and add $9!s(c^{\text{nsp}}(N, g) + 1)/|\text{Aut}(N)|$ to $T_{10,5}^{\text{nsp}}(\text{type}(g))$.
 4. Obtain $F_{10,5}^{\text{nsp}}(\lambda)$ from (15), and set $F_{10,5}(\lambda) \leftarrow F_{10,5}^{\text{sp}}(\lambda) + F_{10,5}^{\text{nsp}}(\lambda)$. Check the exact divisions and nonnegativity.
 5. Set $u_{10,5} \leftarrow \sum_{\lambda \vdash 10} F_{10,5}(\lambda)/z_\lambda$ by integer Burnside accumulation, and $L_{10,5} \leftarrow F_{10,5}(1^{10})$.
 6. Compute ranks 3 and 4 with Algorithm 3 and ranks 0, 1, 2 with (21).
 7. Return $(u_{10,5} + 2 \sum_{r=0}^4 u_{10,r}, L_{10,5} + 2 \sum_{r=0}^4 L_{10,r})$.
-

Theorem 6.1 (Correctness of the combined count). *With complete parent catalogues, their exact automorphism groups, and exact counts of the formulas expressing basis exchange, Algorithm 5 returns the numbers of unlabeled and labeled matroids on ten elements, across all ranks.*

Proof. Corollary 5.3 gives the numbers of fixed sparse paving matroids. Lemma 4.1 and Proposition 2.4 count the invariant extensions that are not sparse paving for each parent of rank five. Equation (14) also includes every coloop extension through the dual parent of rank four. Theorem 3.1, applied to this family, therefore gives its fixed counts for every permutation with a fixed point. Proposition 3.2 gives the full derangement terms, from which the sparse paving terms are subtracted.

The two families are disjoint and exhaustive, so their fixed counts add to $F_{10,5}(\lambda)$ for every cycle type. Burnside's lemma and the identity term give the two counts at rank five. Theorem 3.3 and (21) establish ranks zero through four. Finally, duality is a bijection on both labeled matroids and isomorphism classes, pairing each rank below five with a distinct rank above five. The last line sums these disjoint ranks exactly once. $\square$

7 Implementation and computation

7.1 Implementation

To compute $\text{Aut}(N)$, the implementation backtracks over bijections of the parent ground set. A candidate image must have the same counts of containing subsets of each size and rank. After each assignment, it checks rank preservation on every subset of the assigned domain. Every automorphism passes both tests, while every completed bijection preserves the full rank function; thus the returned group is exact. Its conjugacy classes are formed by explicitly collecting hgh^{-1} for $h \in \text{Aut}(N)$.

The C++ implementation of Algorithm 2 chooses an undecided lower variable with the most clause occurrences. It records assignments on a trail and undoes them after each branch. Algorithm 1 stores vertex sets as bit masks and attempts component decomposition when fewer than 45 vertices remain or the maximum degree is below 5. Otherwise it branches directly.

Persistent processes take parents from a shared queue. For each parent, a process evaluates one representative of every automorphism conjugacy class, with a fresh graph cache for each representative. It returns the parent's contributions to the sums $T_{n,r}(\mu)$ or $T_{10,5}^{\text{nsp}}(\mu)$. These sums and the final reductions use arbitrary-precision integers. The code calls these arrays **moments**; mathematically they are the weighted sums defined above.

The formulas for derangements use one variable per orbit of subsets and no auxiliary variables. Ganak [8] is run in exact mode. For the type 2^5 at ranks four and five, the driver fixes eight variables of highest clause occurrence and sums all 256 resulting cubes, using (13). Completed results are saved so that an interrupted computation can resume. A job that times out contributes no count until it completes.

The source correspondence is:

Part of the algorithm	Source file in the repository
Counting independent sets	<code>src/split_count.cpp</code>
Invariant modular cuts	<code>src/extension_worker.cpp</code>
Completion counts	<code>src/small_extensions.hpp</code>
Counting sparse paving matroids	<code>scripts/count_sparse.py</code>
Basis exchange formulas	<code>scripts/export_symmetry.py</code>
Burnside's lemma and sum over ranks	<code>scripts/count.py</code>

7.2 Results

Table 1 gives the outputs by rank. The full sums are

$$\boxed{u_{10} = 3\,232\,000\,741\,644}, \quad \boxed{L_{10} = 11\,727\,995\,799\,397\,397\,461}. \quad (22)$$

The unlabeled counts at ranks zero through four agree with [4, Table 1]. The value at rank five is a result of the present computation; the full count at this rank has not yet been replicated independently.

The calculation at rank five separates as follows. For each column, the middle row is the contribution of all nonidentity permutations to the Burnside numerator: $\sum_{\lambda \neq 1^{10}} (10!/z_\lambda) F_{10,5}^{\text{sp}}(\lambda)$ or the same sum with superscript nsp. Adding the first two rows and dividing by $10!$ gives the last row.

Table 1: Computed counts on ten elements. Ranks $10 - r$ have the same values.

Rank r	Unlabeled $u_{10,r}$	Labeled $L_{10,r}$
0	1	1
1	10	1 023
2	128	677 546
3	10 037	16 869 404 175
4	4 886 380 924	17 716 560 870 661 325
5	3 222 227 959 444	11 692 562 643 915 909 321

	Sparse paving	Not sparse paving
Identity	9 544 867 699 823 218 639	2 147 694 944 092 690 682
Nonidentity permutations	102 432 886 765 361	155 742 427 712 518
Unlabeled count	2 630 337 889 305	591 890 070 139

Mayhew–Royle [2, Section 9.2] estimated approximately 2.65×10^{12} sparse paving classes of rank five on ten elements. Our sparse paving count is close to that estimate, which is not an exact value against which the computation can be verified.

The complete tables of numbers of fixed matroids are recorded in `results/fixed_terms.json`. Appendix A lists the twelve counts for derangements at rank five.

The recorded full run took 389.16 seconds on 64 physical cores of two AMD EPYC 9354 processors. This includes generating the parents from the empty matroid (3.83 seconds), the sparse paving calculation (124.85 seconds), ranks three and four (6.92 and 76.72 seconds), and the remaining work at rank five (176.47 seconds). Compilation and the separate validation runs are excluded. The C++ programs were compiled with GCC 13.3.0 and `-O3 -march=native -std=c++17`.

7.3 Checks and reproduction

The generated catalogues reproduce all 55 entries indexed by size and rank through nine elements in [2], including 383 172 classes on nine elements. Across sizes zero through nine, all 385 370 canonical rank arrays match the reference catalogue [9]. The rank axioms, absence of isomorphic duplicates, and closure under duality were checked directly. For the 190 214 parents of rank five on nine elements, the orders of the automorphism groups were also checked by a separate program. Relabeling each parent preserves its contribution to every sum $T_{10,5}^{\text{sp}}(\mu)$.

The aggregate algorithm itself reproduces the known values

(n, r)	$(6, 3)$	$(7, 3)$	$(8, 3)$	$(8, 4)$	$(9, 3)$	$(9, 4)$
$u_{n,r}$	38	108	325	940	1 275	190 214.

These cases exercise both the sums over parents and the formulas for derangements. A fresh computation on ten elements with regenerated parents reproduces all 126 numbers of fixed matroids at ranks three, four, and five. The result and check summaries are in `results/counts.json`, `results/validation.json`, and `results/provenance.json`.

The implementation is in the `matroid-count` repository [10], revision `d1aaa1468a01`. From its root, on Linux with Python 3.10+, GCC, Make, and Boost headers, run:

```
python3 scripts/setup.py --audit
make -j4
make audit
make test
python3 scripts/count.py --workdir runs/n10 --jobs 64
python3 scripts/validate.py --parents runs/n10/parents \
    --ten-run runs/n10 --out runs/n10/validation --jobs 8
```

The computed total is written to `runs/n10/total.json`. The generated parent ledgers, formulas, and solver logs remain in that run directory. The source tree contains the compact result summaries.

A Derangement terms at rank five

Each entry in Table 2 is the number of labeled matroids fixed by one permutation of the indicated type. The last column is obtained by subtraction. Its contribution to the unlabeled count is the sum of that column divided, row by row, by z_λ . Types are ordered by increasing number of parts, then in decreasing lexicographic order, with the parts of each partition written in decreasing order.

Table 2: All twelve derangement types in S_{10} at rank five.

Cycle type	All matroids	Sparse paving	Not sparse paving
10	78	67	11
8, 2	637	389	248
7, 3	28	1	27
6, 4	746	132	614
5^2	24 566	21 969	2 597
$6, 2^2$	5 620	1 146	4 474
5, 3, 2	129	4	125
$4^2, 2$	150 791	53 705	97 086
$4, 3^2$	1 022	16	1 006
$4, 2^3$	792 347	34 257	758 090
$3^2, 2^2$	104 620	16 542	88 078
2^5	584 746 603	217 942 377	366 804 226

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